Knowledge Map

What I Know, What's Known, and What's Open

Patrick S. Mahon

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Abstract

A consolidated reference covering everything explored in this research programme: what I understand, what is known (and by whom), what I proved that turned out to be classical, what remains genuinely open, and where the understanding might lead. Not a paper. A map for finding the next problem.

Contents

1 The central question

When does observationally coherent data determine a genuine probability measure, and what additional commitments does this require?

This decomposes into three layers:

1. **Extension:** compatible finitely additive charges on a directed system of Boolean algebras $\to$ σ-additive probability measure.

2. **Realization:** measure on the dual space (Stone space) $\to$ measure on an underlying state space Ω.

3. **Value-definiteness:** probabilistic description $\to$ simultaneous definite values for all observables.

Each layer requires a commitment the previous does not. The algebraic structure of the observation algebra constrains which commitments are available; distributivity is sufficient to make all three available.

2 Layer 1: From charges to measures

2.1 The classical results (all known)

- **Continuity from above:** A finitely additive charge on an algebra is σ-additive iff $E_n \downarrow \varnothing \Rightarrow \ell(E_n) \to 0$ [17, §13, Theorem D].

- **Carathéodory extension:** A σ-additive premeasure on a semiring extends uniquely to a measure on the generated σ-algebra [17, §13], [5, Vol. 1, Ch. 1].

- **Kolmogorov extension:** Compatible finite-dimensional distributions on a product space extend to a unique probability measure, under standard Borel / tightness conditions [4, Ch. 36].

- **Choksi inverse limits:** Compatible measures on an inverse system of compact Hausdorff spaces extend to the inverse limit [9, Theorem 1].

- **Yosida–Hewitt decomposition:** Every bounded finitely additive charge decomposes uniquely as $\ell = \ell_c + \ell_p$ where ℓ_c is σ-additive and ℓ_p is purely finitely additive [45].

- **KVP measurability criterion:** A Boolean algebra $\mathfrak{A}$ admits a strictly positive σ-additive probability iff (i) $\mathfrak{A}$ is Dedekind σ-complete, (ii) $\mathfrak{A}$ is weakly (σ, ∞)-distributive, and (iii) $\mathfrak{A}$ is chargeable (admits a strictly positive finitely additive functional) [13, 391D]. Chargeability is characterized by Kelley's intersection-number condition [13, 391J]. Originally Kantorovich–Vulikh–Pinsker (1950).

- **Conglomerability $\leftrightarrow$ countable additivity:** For precise (linear) previsions, $\mathcal{B}$-conglomerability is equivalent to countable additivity on $\mathcal{B}$ [39].

2.2 What I proved (turned out classical)

Collective exhaustion (CE) in a directed system: defined as $\exists j \geq i$ such that $\ell_j(\pi_{ij}^{-1}(E_n)) \to 0$. Under compatibility, the existential witness is vacuous: $\ell_j(\pi_{ij}^{-1}(E_n)) = \ell_i(E_n)$ for all $j \geq i$. So CE $= \sigma$-additivity of each ℓ_i. This is not a theorem; it is the observation that the definition contains a redundant quantifier.

The directed-system extension theorem (Paper I, Theorem 2.3) is Kolmogorov extension in directed-system language with "sequential common refinements" replacing the usual product structure.

The Lean formalization (0 sorry on the Carathéodory route, 1 sorry for Yosida–Hewitt on the Stone route) is genuine verified infrastructure, even though the theorems are classical.

2.3 What I understand from this

1. The passage from finite additivity to σ-additivity is the critical non-finitary step in probability. Compatibility, normalization, and finite additivity are all first-order / finitary conditions.

2. No first-order condition can force σ-additivity. Immediate from Łoś's theorem: the ultraproduct of Dirac masses on $\mathbb{N}$ is a purely finitely additive charge [20, 12].

3. The finite-cofinite charge on $\mathbb{N}$ is the canonical obstruction: normalized, finitely additive, not σ-additive, and arises as the ultraproduct witness.

4. The Yosida–Hewitt decomposition gives the exact characterization: CE $\Leftrightarrow \ell_p = 0$.

5. The *algebraic* condition for a Boolean algebra to *admit* a strictly positive σ-additive probability is weak (σ, ∞)-distributivity (plus Dedekind σ-completeness and chargeability; KVP 1950, Fremlin 391D). This is a genuine lattice condition on how countable infima interact with arbitrary suprema — the algebraic face of CE.

6. Finite additivity has compactness (inconsistency detected by finite subsets) and absoluteness (depends only on $P[E]$, not on the ambient algebra). σ-additivity has neither. This is the structural reason no finitary framework can see the FA/σA boundary [21].

7. The FA/σA boundary is *pointwise invisible* (Kolmogorov extension: at any fixed finite resolution, PFA marginals are indistinguishable from σ-additive ones) but *asymptotically visible* (YH singularity: PFA marginals eventually leave every fixed smoothness class as resolution increases). The boundary is "felt but not seen": invisible at each snapshot, visible in the trajectory of increasing resolution driven by increasing data.

3 Layer 2: From dual space to realization

3.1 The classical results (all known)

- **Stone representation:** Every Boolean algebra is isomorphic to the clopen algebra of a compact, Hausdorff, totally disconnected space (its Stone space) [23].

- **Charge $\leftrightarrow$ measure on Stone space:** Every finitely additive charge on a Boolean algebra B corresponds to a regular Borel (Baire) measure on its Stone space $\mathrm{St}(B)$ (the space of ultrafilters on B) [13, §326], [17, §53].

- **σ-additivity $\leftrightarrow$ support:** A charge on B is σ-additive iff the corresponding Stone-space measure vanishes on meagre Baire sets, equivalently assigns full mass to countably complete ultrafilters [35, Theorem 10.5.3].

- **Principal ultrafilters:** Every principal ultrafilter is countably complete. The converse holds for $\mathcal{P}(\kappa)$ when κ is below the first measurable cardinal (Ulam's theorem). For non-σ-complete algebras (typical cylinder algebras), countably-complete non-principal ultrafilters can exist without large cardinals.

3.2 What I proved (turned out classical)

Stone measure exists unconditionally: Any compatible charges produce a Baire probability measure $\hat{\mu}$ on $\mathrm{St}(\mathcal{C})$. This is Stone representation + Choksi. Lean-verified (0 sorry).

Descent requires $\hat{\mu}(\mathrm{pure}(\Omega)) = 1$ (where $\mathrm{pure}(\omega) = \{E \in \mathcal{C} : \omega \in E\}$ is the principal ultrafilter at ω)**:** concentration on principal ultrafilters. Under discriminability and the assumption that all countably-complete ultrafilters in the relevant subalgebra are principal (which holds for σ-complete algebras below the first measurable cardinal), this is equivalent to σ-additivity. The support characterization is Rao–Rao 10.5.3; the principal-vs-countably-complete gap is the subtlety.

Realization is unconstrained: Any surjective-projecting $S \subseteq \mathrm{St}(\mathcal{C})$ can serve as Ω. This is the abstractness of Boolean algebras recognized since Stone (1936).

3.3 What I understand from this

1. The Stone construction gives the *unconditional* object: $\hat{\mu}$ always exists. It is the "empirically adequate" object in van Fraassen's sense.
2. Descent to Ω is a *choice*: which ultrafilters count as "realized" is additional structure over the algebra.
3. The three-layer nesting:
$$\mathrm{pure}(\Omega) \subseteq \{\text{countably complete ultrafilters}\} \subseteq \mathrm{St}(\mathcal{C}).$$
σ-additivity gives mass on the middle set (intrinsic). Descent needs mass on the inner set (depends on Ω).
4. The van Fraassen identification (Stone measure = empirical adequacy (EA), descent = realist commitment) has no precedent in the literature. It is a philosophical contribution, not a mathematical one.

4 Layer 3: From probability to value-definiteness

4.1 The classical results (all known)

- **Boolean algebras admit 2-valued homomorphisms:** Ultrafilters on Boolean algebras are exactly the 2-valued homomorphisms. Existence by Zorn. *Stone (1936); every Boolean algebra textbook.*
- **Kochen–Specker:** For the lattice $L(H)$ of closed linear subspaces of a Hilbert space H with $\dim H \geq 3$, no dispersion-free state (2-valued homomorphism) exists [26].
- **Gleason:** For $L(H)$, $\dim H \geq 3$, every σ-additive state is $s(P) = \mathrm{tr}(\rho P)$ [15].
- **Bub–Clifton:** Given a state ρ and a preferred observable R, there is a unique maximal Boolean subalgebra $D(\rho, R) \subseteq L(H)$ on which the state is a mixture of dispersion-free states. [7, 18].
- **Orthomodular lattice (OML) duality:** The category of OMLs is dually equivalent to a category of orthomodular spaces. Principal filters $P(A)$ are part of the dual structure (unlike Stone duality where they are invisible). [30]. Essentially no attention in quantum foundations as of 2026.
- **Colimit generation:** Gluing Boolean algebras via categorical pushout produces orthomodular lattices, generically non-distributive. [16].

4.2 What I proved (classical + one novel framing)

Commensurability theorem: Distributivity is sufficient for value-definite realism (VDR); for $L(H)$, $\dim H \geq 3$, non-distributivity blocks it. Lean-verified (0 sorry, 1 axiom for Kochen–Specker (KS)). *The mathematics is the assembly of Stone + KS + Gleason + Bub–Clifton. Not a new theorem.*

The EA / probabilistic realism (PR) / VDR vocabulary connecting van Fraassen's empirical adequacy to quantum logic via McDonald–Bimbó duality is novel framing. No precedent found.

The coherence filtration $\mathcal{S}(A) \supseteq \mathcal{S}_\sigma(A) \supseteq \mathcal{S}_{\mathrm{df}}(A)$ with three named transitions (pasting, regularity, sharpness) organizes known results. The chain itself is textbook (Pták–Pulmannová, Kalmbach); the transition names and the observation that distributivity collapses the filtration are new packaging.

4.3 What I understand from this

1. **Distributivity controls value-definiteness.** Boolean algebras have ultrafilters (VDR available). Non-Boolean OMLs (dim ≥ 3) do not (KS). The dividing line is distributivity — sufficient but not necessary in full generality (dim 2, horizontal sums).

2. **The algebra can constrain realization in the OML case.** Unlike Boolean Stone duality, the McDonald–Bimbó dual carries $P(A)$ as structure. "Realization is additional structure" in the Boolean case; it is *constrained* structure in the OML case.

3. **The realism debate has different mathematical answers for different algebras.** Boolean: EA, PR, VDR all available; philosophical choice. OML (dim ≥ 3): VDR blocked; PR available (Gleason); EA always available.

4. **The topos programme is more powerful.** Isham–Butterfield / Döring–Isham / Heunen–Landsman–Spitters encode the same structure via spectral presheaves and Bohrification. Our approach is coarser (single dual space, not presheaf over contexts) but connects more directly to van Fraassen.

5 Delay reconstruction

5.1 The classical results

- **Takens:** Generic smooth observable gives an injective delay map for $L \geq 2\dim X + 1$ [43].
- **Sauer–Yorke–Casdagli:** Prevalence-based embedding theorem [38].
- **Generating partitions:** $h(T, P) = h(T)$ iff P generates [41, 27].
- **Grassberger–Procaccia:** Correlation integral estimates D_2 and K_2 (Rényi-2 dimension and entropy) from delay data.
- **False nearest neighbours (FNN):** Embedding dimension selection [25]. Sensitive to noise [36].
- **Ragwitz–Kantz:** Local prediction error minimization for (d, L) selection [34].
- **Botvinick-Greenhouse et al.:** Measure-theoretic Takens via optimal transport [6].

5.2 What I proved (mostly classical)

Reconstruction theorem: Observable σ-algebra = full σ-algebra mod μ iff delay algebra dense in L^2 iff delay map is measure-theoretic embedding. This is Krieger's generating partition theorem [27] in delay-map language.

Predictive factorization, Chapman–Kolmogorov, Koopman–Perron: All classical (Rokhlin, Kolmogorov, Koopman 1931).

Finite-sample convergence rates: Minimax rates $n^{-s/(2s+d)}$ [42]; elbow stopping via Lepski [28]. Not cited in the original work — a significant attribution gap.

5.3 What is genuinely novel (but motivation unclear)

Geometric $\neq$ algebraic reconstruction (2026-05-14): The bridge note claimed $\delta(L) \to 0 \Leftrightarrow (\mu \otimes \mu)(R_L) \to 0$ under a fibre mixing condition. This is **false**: the core identity (Lemma 2.1 of the bridge note) has a disintegration error. The conditional variance $\mathbb{E}[\mathrm{Var}(\mathbf{1}_S \mid \mathcal{O}_L)] = \sum_z p_z \, \mu_z(S)\mu_z(S^c)$ is weighted by p_z, while the collision integral $(1/2)\int_{R_L} |\mathbf{1}_S - \mathbf{1}_{S'}|^2 \, d(\mu \otimes \mu) = \sum_z p_z^2 \, \mu_z(S)\mu_z(S^c)$ is weighted by p_z^2. The bridge note equated them; they are genuinely different.

Skew-product counterexample: $X = A^{\mathbb{Z}} \times B^{\mathbb{Z}}$, $T = \sigma \times \sigma$, $h(a, b) = a_0$, independent iid uniform. Then collision $= 2^{-(L+1)} \to 0$ (fibres shrink in the a-coordinate) but $\delta_L = 1/4$ for all L (the event $\{b_0 = 0\}$ is maximally unresolved). This falsifies even the "easy direction."

What survives: The definitions (collision probability, Rokhlin distance, fibre mixing) are well-posed. The trivial direction $\delta \to 0 \Rightarrow$ collision $\to 0$ holds. The converse and the entropy characterization $\delta \to 0 \Leftrightarrow H_2 \to \infty$ are both dead.

Replacement direction: A short note proving that geometric and algebraic reconstruction are inequivalent, with the skew-product as the central theorem. The open question becomes: what conditions on the factor map make collision $\to 0$ imply $\delta \to 0$?

5.4 What I understand from this

1. The reconstruction problem is *algebraically underdetermined*: the observation algebra does not constrain the realization. Embedding selection (d, L) is necessarily empirical.

2. Every practical reconstruction criterion (FNN, mutual information, Lyapunov convergence, prediction error) implicitly assumes a modelling commitment. Deterministic criteria fail under noise because noise violates the commitment.

3. The Rokhlin distance $\delta(\mathcal{O})$ is the measure-theoretic notion of "how close is the observable σ-algebra to generating everything." It is the Rokhlin metric, named and studied since 1967.

4. Collision probability / Rényi-2 entropy measures the "collision structure" of the fibre decomposition. Grassberger–Procaccia already extracts this from data.

5. The bridge theorem linking collision convergence to σ-algebra generation is false: the disintegration error (p_z vs p_z^2 weighting) means the proposed identity is simply the wrong identity, and the inequivalence of geometric and algebraic reconstruction is known/obvious.

6 Dead ends and lessons

1. **De Finetti via Stone:** Exchangeability forces σ-additivity (Hausdorff moments). The "exotic part" of the Stone measure is empty. Dead end.

2. **Stationary Stone:** The Yosida–Hewitt decomposition commutes with shift-invariance. An observation, not a theorem. The real open question (ergodic theory of the purely finitely additive part) is out of reach.

3. **Dynamics from observation:** "Can the Stone space give rise to dynamics without positing T?" No: the algebra doesn't constrain Ω, so it certainly doesn't constrain structure on Ω.

4. **Paper III (noise thresholds):** Residual autocorrelation as embedding diagnostic = Billings–Voon [3] / Casdagli [8]. FNN noise sensitivity = Rhodes–Morari [36]. Elbow stopping = Lepski [28]. Rediscovery throughout.

5. **"No intermediate condition":** The claim that no condition interpolates between first-order and σ-additivity is false. $\limsup \ell(A_n) < c$ trivially interpolates.

6. **Bridge theorem (fibre mixing):** The claimed equivalence $\delta(L) \to 0 \Leftrightarrow (\mu \otimes \mu)(R_L) \to 0$ under fibre mixing is false. The core identity has a disintegration error: the conditional variance integral is weighted by p_z, the collision integral by p_z^2. The skew-product $X = A^{\mathbb{Z}} \times B^{\mathbb{Z}}$, $h(a, b) = a_0$ falsifies even the easy direction. The replacement observation (geometric $\neq$ algebraic reconstruction) is known to ergodic theorists. Both the positive bridge and the negative note fail the audit. *(2026-06:* a Type-6/4 reframing of the same content — the predictive/geometric *separation* as exposition/vocabulary — was seeded and audited `full`; also **parked**. It *is* the factor-vs-conjugacy distinction, and the delay-map translation is already in the audience's own venues (Botvinick-Greenhouse, *J. Stat. Phys.* 2025, arXiv 2409.08768; functional observability, arXiv 2301.04108). Filed in `covered_leads/relational_reconstruction_separation.md`.*)*

7. **Entropy characterization:** $\delta(L) \to 0 \Leftrightarrow H_2(\nu_L) \to \infty$ depended entirely on the bridge theorem. Dead.

8. **Nonlinear semigroups / Hille operator (Phase 1):** C_n-semigroups on Banach spaces, resolvent from finite difference spectra, max-entropy fitting. Abandoned approach — the semigroup formalism was too heavy for the applied question. The Rose kernel was the lighter successor.

9. **Rose–Takens measure conjugacy:** "Takens embedding + $D_{\mathrm{KL}}(h_{\#}\mu \| \nu) = 0$ implies measure conjugacy." Trivially true by definition: diffeomorphism + pushforward = target measure IS measure conjugacy. Not a theorem.

10. **CGF → Fisher → Landau chain:** Standard information geometry [2]. The CGF-Fisher connection is the definition of Fisher information.

11. **KL = least action (discrete Onsager–Machlup):** Known: [11], [29], large deviations literature.

12. **Fisher clock / entropy production:** Broken detailed balance → entropy production is [40]. The "Fisher clock" framing is poetic but the mathematics is standard nonequilibrium statistical mechanics.

13. **Neutral operator edge law:** The joint edge law $\Gamma = \nu \otimes K$ and its f-divergence structure were absorbed into Paper II (conditional regularity kernel κ_Q). The diagnostic exponents $\eta = 2s$, $\zeta = 2s/(2s + d)$ are the same Lepski/Stone rates already canned in Paper III.

14. **CE via locale theory (Simpson's sublocale):** Conjectured that CE $\Leftrightarrow$ Simpson's smallest measure-1 σ-sublocale is spatial. Dead: δ_U (Dirac at free ultrafilter) gives spatial one-point sublocale but CE fails; compact Hausdorff makes $\Rightarrow$ trivially true for all measures. Non-spatiality is a constructive phenomenon that vanishes classically.

15. **Prediction error $\leftrightarrow$ generation:** Attempted to connect Casdagli's prediction-error method to σ-algebra generation ($\delta(L)$). Self-deflated: Bayes-optimal prediction error IS conditional variance (definitional); the non-trivial direction (prediction of h implies generation) is just the reconstruction theorem (Paper II).

16. **Graded σ-additivity (tail decay):** Proposed a single graded parameter interpolating light tails (σ-additive) $\to$ heavy tails (σ-additive, slow decay) $\to$ PFA (no decay). Contradicted by Schervish–Seidenfeld–Kadane (2021): interior tail behavior and boundary/PFA mass are independent characteristics, not endpoints of one axis. The supremum-over-refinements definition collapses to $\{0, \infty\}$ (recovers only the Halmos binary). The filtration-dependent version is Rényi information dimension under a different name. Taxonomic.

17. **Operational boundary theorem:** Attempted to turn the observation "PFA = Huber contamination" into a theorem with novel content. Two candidate routes killed:
    - *PFA structure improves the contamination floor:* No. At fixed resolution, Kolmogorov extension makes PFA indistinguishable from arbitrary contamination. The floor stays ε^2.
    - *Trajectory $\hat{\varepsilon}(n) \to \varepsilon_0$ as consistent test:* Correct, but known. The mechanism (YH singularity forces PFA marginals out of smooth classes as $k \to \infty$) is the contrapositive of YH decomposition itself. The consistent test is Patra–Sen (2016) / Chen–Gao–Ren (2016).

    What was learned: Kolmogorov extension is the precise mechanism behind pointwise inaccessibility; YH singularity is the mechanism behind asymptotic distinguishability. The boundary is invisible at each snapshot but visible in the trajectory of increasing resolution. Synthesis folded into literature review §2.9. No new theorem.

18. **"Openness as primitive" (six frameworks):** Tested whether frameworks that take indeterminacy or incompleteness as primitive make contact with the FA/σA boundary: Walley (imprecise probability), Abramsky–Brandenburger (sheaf contextuality), Sambin (positive topology), Dzhafarov–Kujala (contextuality by default), Ellerman (partition logic), Gisin/Linnebo–Shapiro (philosophical). All six parked; none produces a new theorem about when σ-additive extension exists. The kills differ: vocabulary-only (Walley — the load-bearing theorem is SSK (1984)); compact/non-compact mismatch (AB, CbD — obstructions witnessed by finite data, while the FA/σA gap is invisible to any finite test); wrong primitives (Sambin — positivity is co-inductive, no measure theory); no measure-theoretic content (Ellerman, Gisin, Linnebo–Shapiro). Details in the reading directions document.

7 What remains genuinely open

Open Problem 7.1 (Measure-free Boolean algebra)**.** Does there exist a Boolean algebra B satisfying all three conditions:

(i) B is non-σ-complete (equivalently, $\mathrm{St}(B)$ is not basically disconnected);

(ii) B is non-atomic (equivalently, $\mathrm{St}(B)$ has no isolated points);

(iii) B is measure-free: B admits no strictly positive σ-additive probability (equivalently, $\mathrm{St}(B)$ carries no strictly positive Radon probability measure)?

Here σ-additivity on a non-σ-complete B is interpreted for disjoint countable families whose join exists in B.

On the clopen algebra of any Stone space, σ-additivity is vacuous by compactness; the real question concerns Radon measures on the Stone space itself (the Borel σ-algebra generated by clopens). Nearest known constructions (Argyros 1983, under CH; Kunen; Fedorchuk) each miss at least one required condition. Plebanek (2024 survey) confirms the exact parameter regime remains open.

RETIRED 2026-06-18 (`/audit full`, prior-art). The question above has a *positive* ZFC answer and is *not* a contribution. The Argyros *pre-Gleason* clopen algebra $\mathrm{Clop}(Y, \mathfrak{T})$ satisfies all three in ZFC — (ii),(iii) are Argyros 1983, and (i) holds by an elementary gap-family witness at $\bar{0}$. But **Gaifman 1964** (PJM 14(1):61–73, Thm 2.2 + property (†)) already exhibits, in ZFC and 19 years prior, an atomless Boolean algebra with no strictly-positive *finitely*-additive measure — strictly stronger. non-σ-complete + atomless are trivial; the only hard ingredient (measure-freeness) is Argyros's published theorem. "Strategy D" was this programme's private name, never field-open; the "likely ZFC-independent" belief was a local misreading (σ-completeness read off the completed/Gleason-cover form, not the base). Retired as a research target; survives only as a one-line Gaifman-1964 citation. See `strategy_d_AUDIT_VERDICT.md`; the `strategy_d_dossier.md` reduction is the (closed) reasoning trail.

Open Problem 7.2 (OML extension)**.** Let A be an orthomodular lattice and $s\colon A \to [0,1]$ a state (i.e. $s(1) = 1$ and $s(a \vee b) = s(a) + s(b)$ whenever $a \perp b$). Let $S_0(A)$ be the McDonald–Bimbó dual space with clopen $\perp$-stable sets $\mathrm{CO}(S_0(A))^\dagger$, and define $\mu(h(a)) = s(a)$ where $h(a) = \{x \in F(A) : a \in x\}$.

(a) Under what conditions on A does μ extend to a σ-additive measure on the Baire σ-algebra of $S_0(A)$?

(b) In particular, does any condition on A strictly weaker than Booleanity suffice?

For $A = L(H)$ with $\dim H \geq 3$, Gleason's theorem gives the extension: every state is $s(P) = \mathrm{tr}(\rho P)$. For general OMLs, Gleason-type results are unavailable and Carathéodory does not apply (the clopens form an OML, not a Boolean algebra).

The structural obstruction is combinatorial. Pták–Pulmannová (1994) prove that if an OML possesses a unital set of subadditive probability measures, it must be Boolean (the "too strong" side). On the "too weak" side, Pták [32] constructs *exotic logics*: σ-orthocomplete OMPs admitting FA states but no σ-additive states, via Greechie pasting — atom identifications across Boolean blocks force contradictory partition-of-unity constraints at the σ-additive level. Each block individually admits σ-additive states; the pasting makes them globally incompatible. The center can be prescribed independently (Theorem 3). Navara [31] sharpens this: regularity (P-regularity, inner regularity w.r.t. finite-dimensional projections), the natural candidate admissibility condition from von Neumann projection lattices (Béaver–Cook 1989), does not force σ-additivity on general OMPs. The gap between too weak and too strong appears structurally robust. Bunce–Wright (1992), Chetcuti–Dvurečenskij (2003/2005), and Voráček–Pták (2023) address adjacent questions without breaking through on the extension problem proper.

Descent axis ($\mathcal{L}_{\mathrm{MO}_2}$ killed 2026-06-10; reframed to a live open problem 2026-06-12; structural dichotomy landed 2026-06-17). The problem splits into two independent axes (survey `oml_onboarding`): an *extension* axis (state $\to$ charge on the full clopen Boolean algebra of $S_0(A)$), governed by the meet-zero/orthogonal gap and *closed* on $L(H)$

— no state extends, by a finite MO_2 construction with infinite-dimensional atoms; and a *descent* axis (concentration on physical points), governed by σ-additivity. The $\mathcal{L}_{\mathrm{MO}_2}$ lead was killed by `/audit full` on concreteness (concrete but trivial — $\prod_n \mathrm{MO}_2$ carries the same bundle — and already characterized by Pták–Pulmannová 1994), *but* the axis was then reframed under Reading 1 to a sharp *uninhabited* open problem: a σ-complete concrete OML with a σ-essential *contextual* state (a compactness failure reconnecting to CE). A 2026-06-17 push *landed a structural dichotomy* (not a verdict): the σ-additive dispersion-free points satisfy $S^{\sigma}_{\mathrm{df}} = S_{\mathrm{df}} \cap \bigcap_{\text{block } B} O_B$ (O_B open), so $\leq \aleph_0$ atomic blocks force $\mu(\text{fakes}) = 0$ (via per-block nullity + countability) and hence *no witness* (Exit B); a witness *necessarily* needs uncountably many infinite atomic blocks, leaving one live construction target (uncountably many atomic blocks on a countable ground set, $|L| = \mathfrak{c}$). [The G_δ-ness of S^{σ}_{df} under $\leq \aleph_0$ blocks is a parallel topological sibling, *not* the cause of conullity.] **Update 2026-06-20:** the "uninhabited open problem" framing is superseded — prior-art (Derr–Williamson 2023, via Maharam 1972 §8) settles it *negatively* in the Polish-representable case (cell empty), leaving open only the non-(topologically-)representable case = the σ-Loomis–Sikorski wall. For the current status see the programme overview (`notes/programme/program_overview.md`, item 4, authoritative; the AFTERLIFE paragraph), the survey `oml_onboarding` (now Reading-1-committed), and `notes/open_questions/reading1_prize_reduction.md`. This reference document does not track that state.

Open Problem 7.3 (Ultralimit representability — reduces to Problem 7.1)**.** Let B be a Boolean algebra, $\mathcal{U}$ a nonprincipal ultrafilter on an index set I, and $(\mu_i)_{i \in I}$ a family of σ-additive probabilities on B. Call a purely finitely additive charge ℓ on B *ultralimit-representable* if $\ell(b) = \lim_{\mathcal{U}} \mu_i(b)$ for all $b \in B$, for some such family and ultrafilter.

For which Boolean algebras B is every purely finitely additive charge on B ultralimit-representable?

Status (2026-05-18): settled in all regimes except Problem 7.1.

- *Atomic B (e.g. $\mathcal{P}(\mathbb{N})$):* Every PFA charge is representable. Ultrafilter charges are realized via Diracs; the rest by Krein–Milman.
- *σ-complete B:* Nikodým convergence forces ultralimits of σ-additive measures to stay σ-additive. No PFA charge arises.
- *Non-σ-complete, non-atomic B:* The canonical decomposition $B \cong (B{\restriction}a) \times (B{\restriction}a^c)$ shows that failure of representability is witnessed by a measure-free direct-product factor — reducing to Problem 7.1.

Literature scoped (2026-05-18): Duanmu–Weiss (2018), Cardona et al. (2025), Fremlin §§326–328 checked; none address this question. Not an independent direction.

Open Problem 7.4 (Extreme PFA shift-invariant charges)**.** Let $T\colon \Omega \to \Omega$ be a measurable transformation on a standard Borel space. What are the extreme points of the convex set of T-invariant purely finitely additive charges on Ω?

Connected to the structure of Banach limits and the Yosida–Hewitt decomposition of invariant means. *Out of reach currently.*

8 The understanding, condensed

One sentence per insight:

1. The passage from coherent charges to probability requires σ-additivity, which is not derivable from finitary conditions.

2. The Stone construction gives an unconditional measure on the dual space; descent to a realization requires a commitment the algebra cannot force.

3. Which ultrafilters are "realized" is additional structure in the Boolean case and constrained structure in the OML case.

4. Distributivity is sufficient for value-definiteness; non-distributivity blocks it for $L(H)$, $\dim \geq 3$, but not for all non-distributive OMLs (e.g. $L(\mathbb{C}^2)$, horizontal sums).

5. The realism/empiricism boundary is algebraic: van Fraassen's empirical adequacy is the unconditional level; value-definite realism requires the algebra to admit dispersion-free states (guaranteed by distributivity, blocked by KS for $L(H)$, $\dim \geq 3$).

6. Delay reconstruction is algebraically underdetermined: embedding selection is necessarily empirical.

7. Geometric reconstruction (collision $\to 0$) and algebraic reconstruction ($\delta \to 0$) are genuinely inequivalent. The bridge theorem claiming their equivalence under fibre mixing was false (disintegration error in the core identity). The skew-product counterexample is definitive.

8. Every practical reconstruction diagnostic implicitly assumes a modelling commitment; noise violates deterministic commitments.

9. Audit before you draft.

9 References by subject

Measure extension and charges

- Halmos, *Measure Theory*, Van Nostrand, 1950. §13 (extension), §53–54 (Boolean algebras). [17]

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